

STAC67: Regression Analysis

Lecture 7

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University of Toronto Scarborough

Reminder: Assignment 1 due Friday

Due Friday 2 October, 23:59, on Crowdmark. **Late submissions are not accepted.**

Submitting on Crowdmark

1. Each question has its own upload slot: put the pages for each question in its slot.
2. Questions 1 and 7: the knitted R Markdown output, with your code and its output visible. If both are in one file, upload that file to both slots.
3. Questions 2 to 6: typed or handwritten. Scans must be clearly legible.
4. Once everything is uploaded, press **Submit** (uploading alone does not submit).
 - Show your work: an answer without justification receives no credit.

Leave yourself time for scanning and uploading: do not start at 23:45.

Review of Lecture 6

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Today: all of this on a real dataset and the natural question

Are the assumptions behind these intervals and tests actually true?

Case study: COVID-19 in Chicago

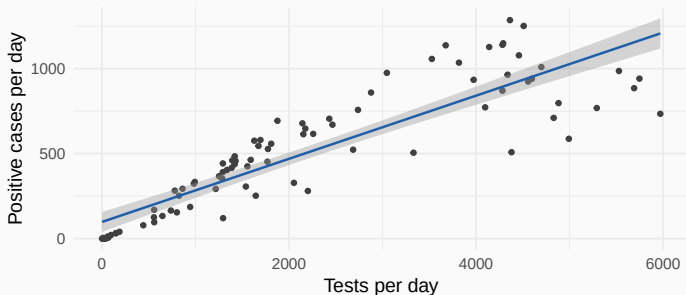
Daily counts for the city of Chicago in spring 2020, $n = 89$ days: the number of tests performed and the number of positive cases (file COVID-19.csv).

1. Draw a scatter plot of the data.
2. Find the correlation coefficient, and test whether the number of tests and the number of cases are linearly related.
3. Regress cases on tests: find the estimated intercept and slope, and the estimate of σ .
4. Give a 98% confidence interval for the slope.
5. Predict the number of cases on a day with 10, 100, 1000 and 5000 tests.
6. Give 95% confidence intervals for the mean number of cases at these values, and 95% prediction intervals.

1. Scatter plot

```
library(ggplot2)
Covid <- read.csv("COVID-19.csv")

ggplot(Covid, aes(Tests, Cases)) +
  geom_point(size = 0.9, colour = "grey25") +
  geom_smooth(method = "lm", colour = "#1F5FA9", linewidth = 0.6) +
  labs(x = "Tests per day", y = "Positive cases per day") +
  theme_minimal(base_size = 10)
```



What do you see? Is a straight line a reasonable summary of these points?

2. Correlation

```
cor.test(Covid$Tests, Covid$Cases)

##
## Pearson's product-moment correlation
##
## data: Covid$Tests and Covid$Cases
## t = 17.405, df = 87, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.8244944 0.9206675
## sample estimates:
##          cor
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Keep the value $t = 17.405$ in mind: the regression on the next slide gives the slope exactly the same t value.

3. The fitted regression

```
fit <- lm(Cases ~ Tests, data = Covid)
summary(fit)
```

```
##
## Call:
## lm(formula = Cases ~ Tests, data = Covid)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -472.73  -99.45    1.19   114.42   376.91
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  97.77689    29.54996   3.309  0.00136 **
## Tests        0.18572     0.01067  17.405 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 173.8 on 87 degrees of freedom
## Multiple R-squared:  0.7769, Adjusted R-squared:  0.7743
## F-statistic: 302.9 on 1 and 87 DF, p-value: < 2.2e-16
```

Fitted line: $\hat{Y} =$

$\hat{\sigma} =$

Interpretation of $\hat{\beta}_1$:

4. A 98% confidence interval for the slope

```
confint(fit, level = 0.98)
```

```
##              1 %              99 %  
## (Intercept) 27.7441770 167.8096011  
## Tests      0.1604334   0.2110128
```

By hand, with $t(0.99; 87) = 2.370$:

$$\hat{\beta}_1 \pm t(0.99; 87) SE(\hat{\beta}_1) =$$

Interpretation

5 and 6. Predictions, and two kinds of interval

```
new.data <- data.frame(Tests = c(10, 100, 1000, 5000))  
predict(fit, new.data, interval = "confidence")
```

```
##           fit          lwr          upr  
## 1    99.63412   41.06609   158.2022  
## 2   116.34920   59.25869   173.4397  
## 3   283.49999  239.32699   327.6730  
## 4 1026.39241  956.00421 1096.7806
```

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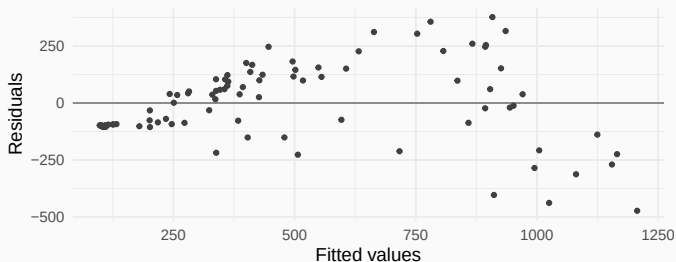
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The arithmetic is right. It is the model that is wrong. The residuals will tell us where.

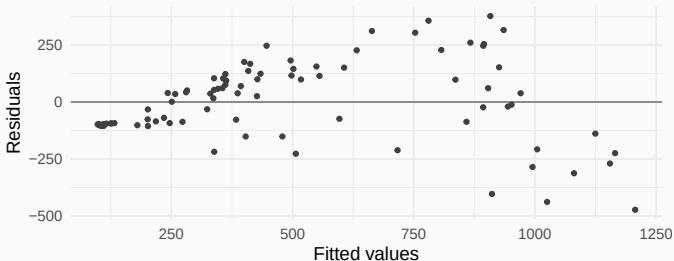
Look at the residuals

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res <- data.frame(fitted = fitted(fit), resid = resid(fit))
ggplot(res, aes(fitted, resid)) +
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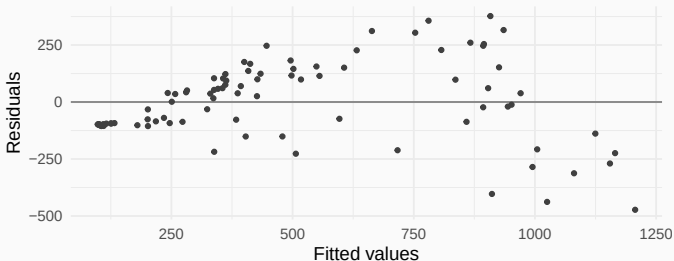
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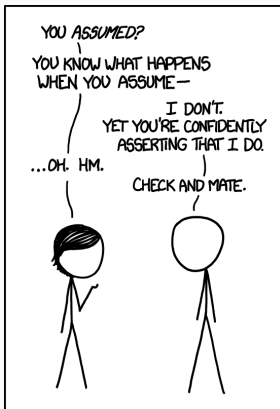


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And the 89 observations are **consecutive days**: one day's error is likely to be close to the next day's.

Diagnostics and Remedial Measures

You assumed?



Every interval and test since Lecture 5 has rested on assumptions about the errors. Today we write them down, and start checking them.

xkcd.com/1339 (CC BY-NC 2.5)

Model assumptions

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, \dots, n$$

Key assumptions of the simple linear regression model:

- (L) **Linearity**: the mean of Y is linear in X , $E(Y | X) = \beta_0 + \beta_1 X$
- (I) **Independence**: the errors are independent, or at least uncorrelated:
 $Cov(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$
- (N) **Normality**: $\epsilon_i \sim N(0, \sigma^2)$
- (E) **Equal variance**: $Var(\epsilon_i) = \sigma^2$ for every i

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Inference relies on the model being true. If an assumption fails, the estimates can be **biased**, and the standard errors, intervals and tests can be **wrong**.

What each result needs

Result	Assumptions used
$\hat{\beta}_0, \hat{\beta}_1$ unbiased	L
$Var(\hat{\beta}_1) = \sigma^2 / S_{xx}$, Gauss-Markov (BLUE)	L, I, E
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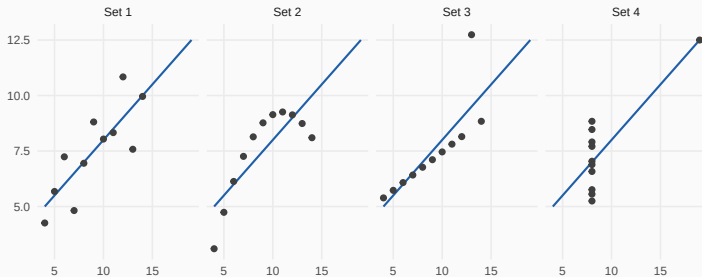
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We cannot observe the errors ϵ_i to perform statistical inference on them, so we check the assumptions with their approximations instead, the residuals $e_i = Y_i - \hat{Y}_i$, mostly by plotting them.

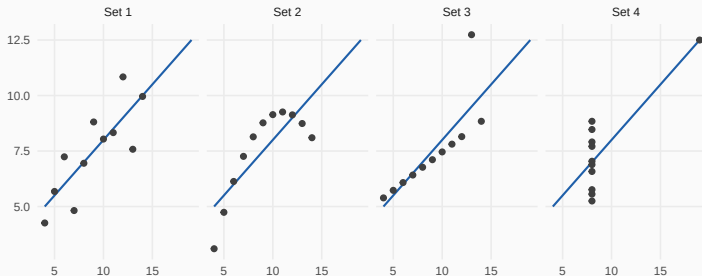
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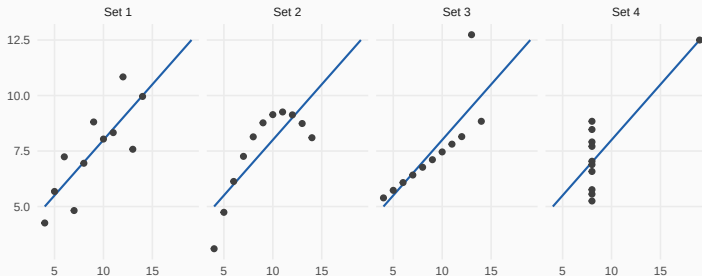
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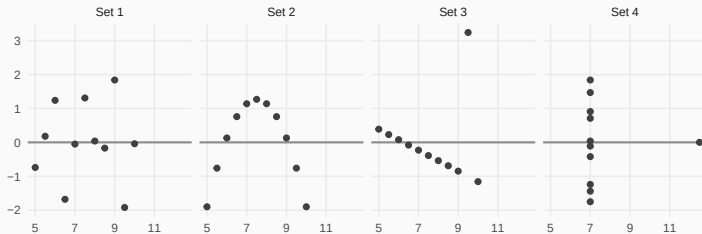


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The regression output cannot tell these four apart. A plot can.

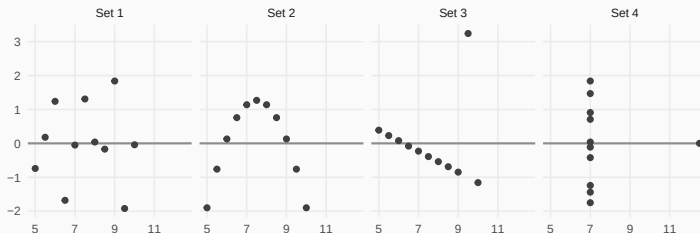
Residual plots

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- Set 1: no pattern. This is what a residual plot should look like.
 - Set 2: a clear curve. **Linearity** fails.
 - Set 3: one outlier, while the other ten points lie exactly on a line.
 - Set 4: one point alone determines the slope, yet its residual is zero.
- ⚠️ A residual plot can miss a point with this much influence.

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Next lecture: a closer look at the residuals, formal checks of each assumption, and what to do when one fails.